

SHUZE WANG

shuzwang@umich.edu | shuzwang.github.io | Boston, MA

SUMMARY

Computational biologist and bioinformatics scientist (Ph.D., University of Michigan) with over seven years applying machine learning and software engineering to biological problems — single-cell multi-omics, spatial transcriptomics, regulatory genomics, and deep learning for biological imaging. Proven record of building analysis pipelines and computational tools adopted by wet-lab biologists, fine-tuning deep learning and foundation models, and publishing ML-for-biology and cancer genomics research in Nature Communications, Blood Cancer Discovery, Developmental Cell, Genome Research, and PNAS. Currently a Bioinformatics Specialist II in the Molecular Pathology Unit at Massachusetts General Hospital and the Broad Institute, applying computation to clinical and translational cancer biology.

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** PyTorch, PyTorch Lightning; training, fine-tuning, and interpretation of deep learning models; sequence-based deep learning; image segmentation; tabular-data foundation model fine-tuning.
- **Programming & Software Engineering:** Python, R/RShiny, Julia, MATLAB, C, C#, Java, CSS, HTML; Git/GitHub; Unix/shell/bash; high-performance computing (HPC); LaTeX.
- **Bioinformatics & Genomics:** single-cell RNA-seq, single-cell ATAC-seq, spatial transcriptomics, multi-omics integration; whole-exome sequencing and variant analysis; RNA-seq, ChIP-seq, ATAC-seq, DisP-seq, WGBS; 16S and metagenomic sequencing.
- **Biological Databases:** NCBI, Ensembl, UCSC; GO, KEGG, Reactome, MSigDB; GEO, ENCODE; TCGA, DepMap.
- **Design & Statistics:** Adobe Illustrator, Adobe Photoshop; SAS, SPSS, Stata.

PROFESSIONAL EXPERIENCE

Bioinformatics Specialist II — Molecular Pathology Unit

Dec 2025 – Present

Bioinformatics Specialist I — Molecular Pathology Unit

Jun 2023 – Dec 2025

Massachusetts General Hospital & Broad Institute, Boston, MA

- Develop and maintain production bioinformatics pipelines spanning genomics, epigenomics, transcriptomics, and metagenomics for clinical diagnostic and translational research workflows.
- Analyze high-throughput sequencing data, delivering interpretable results to pathologists and lab scientists.
- Contribute computational analysis to peer-reviewed manuscripts, including cancer studies identifying targetable therapeutic vulnerabilities.

Graduate Student Research Assistant

May 2019 – April 2023

University of Michigan, Dept. of Computational Medicine & Bioinformatics | Advisors: Joerg Waldhaus, Jie Liu

- Built and shipped computational workflows for single-cell multi-omics analysis (scRNA-seq, scATAC-seq) adopted by wet-lab collaborators for downstream biological interpretation (*first-author, Genome Research*).
- Developed a novel algorithm to spatially reconstruct the 2D and 3D structure of the mouse cochlea from multiple scRNA-seq datasets (*first-author, Developmental Cell*).
- Trained deep learning models in PyTorch for nuclei and transcript segmentation from in situ hybridization imaging.
- Analyzed single-cell multi-omics (RNA + ATAC) from isogenic and CHARGE-syndrome patient samples to study neural tube patterning.

Research Assistant

September 2017 – June 2018

University of Michigan, Dept. of Computational Medicine & Bioinformatics | Advisors: Stephen C. J. Parker, Jacob Kitzman

- Designed and implemented a computational pipeline for ATACme-seq, a high-throughput assay measuring chromatin accessibility and DNA methylation simultaneously.
- Validated ATACme-seq by benchmarking against whole-genome bisulfite sequencing; performed chromatin-state enrichment, gene set enrichment, and motif analysis.

Research Assistant

January 2017 – September 2017

University of Michigan, Dept. of Computational Medicine & Bioinformatics | Advisor: Maureen A. Sartor

- Developed distance-based methods to define enhancer loci from ChIP-seq datasets and benchmarking methods to select optimal definitions.
- Curated and maintained gene set databases (KEGG, Reactome, CTD, Pfam), including schema-level updates for existing analysis packages.

EDUCATION

Ph.D., Bioinformatics

University of Michigan, Ann Arbor | GPA 3.98

Aug 2018 – Aug 2023

M.S., Electrical Engineering & Computer Science

University of Michigan, Ann Arbor | GPA 3.98

Aug 2020 – Apr 2022

M.S., Bioinformatics

University of Michigan, Ann Arbor | GPA 4.00

Aug 2016 – Apr 2018

B.M., Electronic Commerce

Central University of Finance & Economics, China | GPA 3.5

Aug 2012 – Jul 2016

SELECTED PUBLICATIONS

- Duffy, J., Rider, A., Mullen, G., Biral, L., Newton, G., **Wang, S.**, ... Louissaint, A. Plasmacytic lineage unifies plasmablastic and ALK-positive large B-cell lymphomas and reveals targetable vulnerabilities in CD19-negative large B-cell lymphomas. *Blood Cancer Discovery* (under revision).
- Wei, Y., Sánchez AC L., Mathavarajah, S., Antelo, D., **Wang, S.**, ... Rivera, M., Rheinbay, E., Langenau, D. MYOD1 mutation drives cancer stem cell pathways and therapy-resistance in spindle cell/sclerosing rhabdomyosarcoma. *Nature Communications* (under revision).
- Chakraborty, S., **Wang, S.**, Ruhala, J., Mehling, H., Liu, J., Waldhaus, J. Retinoic acid receptor assembly dynamics governs dual functions in cochlear organogenesis. *PNAS* 122(26) e2426739122, 2025.
- Waldhaus, J., **Wang, S.**, Welch, CW. Unique cell type-specific signaling patterns define cholesteatoma. *Otology & Neurotology* 46(7):e285–e292, 2025.
- Wang, S.** (first author), Chakraborty, S., Fu, Y., Lee, M. P., Liu, J., Waldhaus, J. 3D reconstruction of the mouse cochlea from scRNA-seq data suggests morphogen-based principles in apex-to-base specification. *Developmental Cell*, 2024.
- Zhao, N., **Wang, S.**, Huang, Q., Dong, S., Boyle, P. A. Explain-seq: an end-to-end pipeline from training to interpretation of sequence-based deep learning models. *bioRxiv*, 2023.
- Wang, S.** (first author), Lee, P. M., Jones, S., Liu, J., Waldhaus, J. Mapping the regulatory landscape of auditory hair cells from single-cell multi-omics data. *Genome Research* 31(10), 1885–1899, 2021.
- Qin, T., Lee, C., Cavalcante, R. G., Orchard, P., Yao, H., Zhang, H., **Wang, S.**, Patil, S., Boyle, A. P., Sartor, M. A. Comprehensive enhancer-target gene assignments improve gene set level interpretation of genome-wide regulatory data. *Genome Biology* 23, 105, 2022.
- Lee, T. C., Cavalcante, G. R., Lee, C., Qin, T., Patil, S., **Wang, S.**, Tsai, T. Y. Z., Boyle, P. A., Sartor, A. M. Poly-Enrich: count-based methods for gene set enrichment testing with genomic regions. *NAR Genomics and Bioinformatics* 2(1), 2020.
- Wang, H., **Wang, S.**, Yi, X., Tao, Y., Qian, H., Jia, P., Chen, Y., Sun, Y. Estimate of ischemic stroke prevalence according to a novel 4-tiered classification of left ventricular hypertrophy. *Annals of Medicine* 50(6), 519–528, 2018.
- Wang, H., Sun, Y., **Wang, S.**, Qian, H., Jia, P., Chen, Y., Li, Z., Zhang, L. Body adiposity index, lipid accumulation product, and cardiometabolic index reveal the contribution of adiposity phenotypes to hyperuricemia risk in a Chinese rural population. *Clinical Rheumatology* 37(8), 2221–2231, 2018.

SELECTED TALKS

- Wang, S.** et al. Opposing gradients of retinoic acid and sonic hedgehog specify tonotopic identity in the mouse cochlea. 30th Intelligent Systems for Molecular Biology (ISMB), Madison, WI, 2022.
- Wang, S.** et al. Molecular mechanism conferring spatial identity during cochlear duct extension. 29th ISMB (2021) and 12th ACM-BCB (2021).
- Wang, S.** et al. Mapping the regulatory landscape of auditory hair cells from single-cell multi-omics data. 14th Great Lakes Bioinformatics (GLBIO) conference, 2021.
- Wang, S.** et al. Single-cell chromatin accessibility delineates cellular identities of the neonatal organ of Corti. 28th ISMB, 2020.

TEACHING

- **Graduate Student Instructor, University of Michigan:** BIOINF 527 (Introduction to Bioinformatics & Computational Biology), Fall 2022; BIOINF 524 (Foundations for Bioinformatics), Winter 2023.